

DSC 291: Learning Theory - PA8

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Part A: Choosing Regularization by Validation

1. Validation selection for a finite random candidate set.

Claim: $E_V L_D(h_{\hat{k}}) \leq \min_{k \in [K]} L_D(h_k) + \sqrt{\frac{\log(2K)}{2n_V}}$

Proof:

Let $L_D(h)$ be the true risk and $L_V(h)$ be the empirical risk on validation set V . Since the predictors h_k are fixed and independent of V , for each $k \in [K]$, the expected value of $L_V(h_k)$ over V is exactly the true risk $L_D(h_k)$.

Let us define the deviation for each candidate as $Z_k = L_D(h_k) - L_V(h_k)$. Since the loss function is bounded, $\ell(w, z) \in [0, 1]$, both the true risk and the empirical risk are bounded in $[0, 1]$. Therefore, the difference Z_k has zero mean and, by Hoeffding's lemma for bounded variables, is sub-Gaussian with variance proxy $1/(4n_V)$. This means for any $s > 0$:

$$E_V[\exp(sZ_k)] \leq \exp\left(\frac{s^2}{8n_V}\right)$$

We are interested in bounding the expected maximum deviation, so let $Z = \max_{k \in [K]} Z_k$. Using Jensen's inequality (since the exponential function is convex) and the fact that the maximum of non-negative elements is bounded by their sum, we have:

$$\exp(sE_V[Z]) \leq E_V[\exp(sZ)] = E_V\left[\max_{k \in [K]} \exp(sZ_k)\right] \leq \sum_{k=1}^K E_V[\exp(sZ_k)] \leq K \exp\left(\frac{s^2}{8n_V}\right)$$

Taking the natural logarithm of both sides and dividing by $s > 0$:

$$E_V[Z] \leq \frac{\log K}{s} + \frac{s}{8n_V}$$

To minimize this bound, we plug in $s = \sqrt{8n_V \log K}$, which yields:

$$E_V[Z] \leq \sqrt{\frac{\log K}{2n_V}}$$

Now, let $k^* \in \arg \min_{k \in [K]} L_D(h_k)$ be the index of the predictor with the minimum true risk. For the empirically selected predictor $\hat{k}$, we can bound its true risk:

$$L_D(h_{\hat{k}}) = L_V(h_{\hat{k}}) + (L_D(h_{\hat{k}}) - L_V(h_{\hat{k}})) \leq L_V(h_{\hat{k}}) + Z$$

Because $\hat{k}$ was chosen specifically to minimize the empirical validation risk L_V , we know that $L_V(h_{\hat{k}}) \leq L_V(h_{k^*})$. Thus:

$$L_V(h_{\hat{k}}) + Z \leq L_V(h_{k^*}) + Z$$

Taking the expectation over the random validation set V :

$$E_V[L_D(h_{\hat{k}})] \leq E_V[L_V(h_{k^*})] + E_V[Z]$$

Since h_{k^*} is fixed independently of V , its expected empirical risk is its true risk. Substituting our bound for $E_V[Z]$:

$$E_V[L_D(h_{\hat{k}})] \leq L_D(h_{k^*}) + \sqrt{\frac{\log K}{2n_V}} = \min_{k \in [K]} L_D(h_k) + \sqrt{\frac{\log K}{2n_V}}$$

Since $K \leq 2K$, this implies the slightly looser bound given in the claim:

$$E_V[L_D(h_{\hat{k}})] \leq \min_{k \in [K]} L_D(h_k) + \sqrt{\frac{\log(2K)}{2n_V}}$$

2. Oracle inequality over a grid.

Claim: $E L_D(h_{\hat{\lambda}}) \leq \min_{\lambda \in \Lambda} \inf_{u \in \mathcal{W}} \left\{ L_D(u) + \lambda \Psi(u) + \frac{2G^2}{\lambda a n_T} \right\} + \sqrt{\frac{\log(2K)}{2n_V}}$

Proof:

From the setup, the sample is split into independent training and validation sets, T and V . Conditionally on the training sample T , the trained rules h_{λ} for $\lambda \in \Lambda$ are fixed and entirely independent of V . Therefore, we can apply the result from Part 1 directly with $K = |\Lambda|$:

$$E_V[L_D(h_{\hat{\lambda}}) \mid T] \leq \min_{\lambda \in \Lambda} L_D(h_{\lambda}) + \sqrt{\frac{\log(2K)}{2n_V}}$$

Now, taking the expectation over the random training sample T :

$$\mathbb{E}[L_D(h_{\hat{\lambda}})] = \mathbb{E}_T[\mathbb{E}_V[L_D(h_{\hat{\lambda}}) \mid T]] \leq \mathbb{E}_T \left[\min_{\lambda \in \Lambda} L_D(h_{\lambda}) \right] + \sqrt{\frac{\log(2K)}{2n_V}}$$

By Jensen's inequality (or simply exchanging the expectation and the minimum, since the minimum of expectations is $\geq$ the expectation of the minimum), we have:

$$\mathbb{E}_T \left[\min_{\lambda \in \Lambda} L_D(h_{\lambda}) \right] \leq \min_{\lambda \in \Lambda} \mathbb{E}_T[L_D(h_{\lambda})]$$

For any fixed $\lambda \in \Lambda$, h_{λ} is the Regularized Empirical Risk Minimizer (RERM) trained on T . By the standard Learning from Stable RERM theorem, since ℓ is G -Lipschitz and Ψ is α -strongly convex, for any arbitrary comparator $u \in \mathcal{W}$:

$$\mathbb{E}_T[L_D(h_{\lambda})] \leq L_D(u) + \lambda \Psi(u) + \frac{2G^2}{\lambda \alpha n_T}$$

Because this holds for any $u \in \mathcal{W}$, we can take the infimum over u :

$$\mathbb{E}_T[L_D(h_{\lambda})] \leq \inf_{u \in \mathcal{W}} \left\{ L_D(u) + \lambda \Psi(u) + \frac{2G^2}{\lambda \alpha n_T} \right\}$$

Substituting this back into our expectation bound and taking the minimum over $\lambda \in \Lambda$:

$$\min_{\lambda \in \Lambda} \mathbb{E}_T[L_D(h_{\lambda})] \leq \min_{\lambda \in \Lambda} \inf_{u \in \mathcal{W}} \left\{ L_D(u) + \lambda \Psi(u) + \frac{2G^2}{\lambda \alpha n_T} \right\}$$

Combining all steps yields the desired oracle inequality:

$$\mathbb{E}L_D(h_{\hat{\lambda}}) \leq \min_{\lambda \in \Lambda} \inf_{u \in \mathcal{W}} \left\{ L_D(u) + \lambda \Psi(u) + \frac{2G^2}{\lambda \alpha n_T} \right\} + \sqrt{\frac{\log(2K)}{2n_V}}$$

Explanation: Selecting among K trained RERM rules means we evaluate each rule on the validation set to pick the best one. However, the empirical validation loss is only a random estimate of the true risk. The union bound over all K candidates forces us to pay a

complexity penalty of $\sqrt{\frac{\log K}{n_V}}$ (or $\sqrt{\frac{\log(2K)}{2n_V}}$ specifically here). This term acts as a uniform

convergence bound, ensuring that our empirical estimate of the validation loss faithfully tracks the true risk across all candidate rules, preventing us from "overfitting" the validation set.

3. Adapting to an unknown comparator scale.

Claim: $EL_D(h_{\hat{\lambda}}) \leq L_D(u) + \frac{5\sqrt{2}}{2} \frac{G\sqrt{\Psi(u)}}{\sqrt{an_T}} + \sqrt{\frac{\log(2K)}{2n_V}}$

Proof:

Let $u \in W$ be a comparator such that $B_{\min}^2 \leq \Psi(u) \leq B_{\max}^2$. Let $a = \Psi(u) > 0$. The two-term expression we want to optimize is:

$$f(\lambda) = \lambda a + \frac{2G^2}{\lambda a n_T}$$

If we could choose a continuous value of λ , taking the derivative with respect to λ and setting it to zero gives the optimal continuous choice:

$$\lambda_{\text{opt}}(a) = \sqrt{\frac{2G^2}{a n_T}}$$

At this optimal value, the two terms are perfectly balanced:

$$\lambda_{\text{opt}}(a)a = \sqrt{\frac{2G^2 a}{a n_T}} \quad \text{and} \quad \frac{2G^2}{\lambda_{\text{opt}}(a)a n_T} = \sqrt{\frac{2G^2 a}{a n_T}}$$

So the minimum continuous value is $2\sqrt{\frac{2G^2 a}{a n_T}}$.

However, we are restricted to a finite grid Λ . The problem states that the grid has the property that there exists some $\lambda \in \Lambda$ satisfying:

$$\frac{1}{2}\lambda_{\text{opt}}(a) \leq \lambda \leq 2\lambda_{\text{opt}}(a)$$

We can represent this chosen grid point as $\lambda = c\lambda_{\text{opt}}(a)$ for some constant $c \in [1/2, 2]$.

Substituting $\lambda = c\lambda_{\text{opt}}(a)$ into our two-term expression:

$$\lambda\Psi(u) + \frac{2G^2}{\lambda a n_T} = c\lambda_{\text{opt}}(a)a + \frac{2G^2}{c\lambda_{\text{opt}}(a)a n_T} = c\sqrt{\frac{2G^2 a}{a n_T}} + \frac{1}{c}\sqrt{\frac{2G^2 a}{a n_T}} = \left(c + \frac{1}{c}\right)\sqrt{\frac{2G^2 a}{a n_T}}$$

We need to find the worst-case penalty for this discretization over the interval $c \in [1/2, 2]$.

The function $g(c) = c + \frac{1}{c}$ is strictly convex for $c > 0$. Therefore, its maximum on any closed interval must occur at one of the boundaries. Checking the boundaries:

- For $c = 1/2$, $g(1/2) = 1/2 + 2 = 2.5 = 5/2$.
- For $c = 2$, $g(2) = 2 + 1/2 = 2.5 = 5/2$.

Thus, for our chosen grid point λ , the expression is bounded by:

$$\lambda \Psi(u) + \frac{2G^2}{\lambda a n_T} \leq \frac{5}{2} \sqrt{\frac{2G^2 a}{a n_T}} = \frac{5}{2} \sqrt{2} \frac{G \sqrt{\Psi(u)}}{\sqrt{a n_T}} = \frac{5\sqrt{2}}{2} \frac{G \sqrt{\Psi(u)}}{\sqrt{a n_T}}$$

Using the oracle inequality from Part 2, which states that the expected loss is bounded by the minimum over all $\lambda \in \Lambda$, we can safely upper bound this minimum by evaluating it at the specific λ we just analyzed. This yields the final bound:

$$EL_D(h_{\hat{\lambda}}) \leq L_D(u) + \frac{5\sqrt{2}}{2} \frac{G \sqrt{\Psi(u)}}{\sqrt{a n_T}} + \sqrt{\frac{\log(2K)}{2n_V}}$$

4. A dyadic grid and the price of tuning.

Claim: A dyadic grid of size $K = \lceil \log_2(B_{\max}/B_{\min}) \rceil + 1$ satisfies the condition, leading to the simplified guarantee:

$$EL_D(h_{\hat{\lambda}}) \leq L_D(u) + 5 \frac{G \sqrt{\Psi(u)}}{\sqrt{a n}} + \sqrt{\frac{\log(2K)}{n}}$$

Proof:

The required grid must cover the full range of optimal λ values for any comparator norm bounded by $a \in [B_{\min}^2, B_{\max}^2]$. Since $\lambda_{\text{opt}}(a)$ is inversely proportional to $\sqrt{a}$, the extreme values of λ we need to cover are:

$$\lambda_{\max} = \lambda_{\text{opt}}(B_{\min}^2) = \frac{\sqrt{2}G}{B_{\min} \sqrt{a n_T}}, \quad \lambda_{\min} = \lambda_{\text{opt}}(B_{\max}^2) = \frac{\sqrt{2}G}{B_{\max} \sqrt{a n_T}}$$

A simple way to satisfy the requirement that a grid point λ exists within a factor of 2 of $\lambda_{\text{opt}}(a)$ is to use a dyadic grid (powers of 2). We can construct the grid starting from $\lambda_{\max}$ and halving it repeatedly:

$$\Lambda = \{\lambda_{\max} 2^{-k} : k = 0, 1, \dots, K-1\}$$

To ensure that even the smallest possible optimal value $\lambda_{\min}$ has a grid point λ satisfying $\lambda \leq 2\lambda_{\min}$, the smallest point in our grid must be at most $2\lambda_{\min}$:

$$\lambda_{\max} 2^{-(K-1)} \leq 2\lambda_{\min} \implies \lambda_{\max} 2^{-K} \leq \lambda_{\min} \implies 2^K \geq \frac{\lambda_{\max}}{\lambda_{\min}}$$

Substituting our definitions for $\lambda_{\max}$ and $\lambda_{\min}$:

$$2^K \geq \frac{\sqrt{2}G/(B_{\min}\sqrt{an_T})}{\sqrt{2}G/(B_{\max}\sqrt{an_T})} = \frac{B_{\max}}{B_{\min}}$$

Thus, taking the base-2 logarithm, the number of grid points required is exactly $K = \lceil \log_2(B_{\max}/B_{\min}) \rceil + 1$ (which is $O(\log(B_{\max}/B_{\min}))$).

Now, assuming n is even and setting the data split to be $n_T = n_V = \frac{n}{2}$, we substitute this into our bound from Part 3:

$$\frac{5\sqrt{2}}{2} \frac{G\sqrt{\Psi(u)}}{\sqrt{a(n/2)}} = \frac{5\sqrt{2}}{2} \frac{\sqrt{2}G\sqrt{\Psi(u)}}{\sqrt{an}} = \frac{10}{2} \frac{G\sqrt{\Psi(u)}}{\sqrt{an}} = 5 \frac{G\sqrt{\Psi(u)}}{\sqrt{an}}$$

And for the validation term:

$$\sqrt{\frac{\log(2K)}{2(n/2)}} = \sqrt{\frac{\log(2K)}{n}}$$

This gives the simplified guarantee:

$$EL_D(h_{\hat{\lambda}}) \leq L_D(u) + 5 \frac{G\sqrt{\Psi(u)}}{\sqrt{an}} + \sqrt{\frac{\log(2K)}{n}}$$

Identification of terms:

- RERM learning term:** $5 \frac{G\sqrt{\Psi(u)}}{\sqrt{an}}$. This term represents the core statistical error of learning with an appropriately tuned regularized empirical risk minimizer. Notice that it is penalized (inflated by a constant factor) because we are forced to use only half the data for training and because our grid discretization is slightly suboptimal.
- Validation-selection term:** $\sqrt{\frac{\log(2K)}{n}}$. This bounds the error from evaluating K different candidate rules on the finite validation set. It is the cost of searching for the best parameter.
- Additional term caused by not knowing comparator scale:** Because we do not know the correct scale $\Psi(u)$ in advance, we cannot simply fix an optimal λ . We are forced to split our data (halving the effective n) and tune over a grid. This directly introduces the entire validation-selection penalty $\sqrt{\frac{\log(2K)}{n}}$, and simultaneously inflates the RERM learning term from its ideal $2\sqrt{2}$ factor to 5.

Part B: Approximate RERM and Optimization Error

1. Approximate minimizers are close to exact minimizers.

Claim: $\|\tilde{w}_S - w_S\| \leq \sqrt{\frac{2\eta}{\lambda\alpha}}$

Proof:

Consider the regularized objective function $F_S(w) = L_S(w) + \lambda\Psi(w)$. This function is the sum of a convex empirical loss $L_S(w)$ and a regularizer $\lambda\Psi(w)$. Because Ψ is assumed to be α -strongly convex with respect to the norm $\|\cdot\|$, scaling it by $\lambda > 0$ means that $\lambda\Psi(w)$ is $\lambda\alpha$ -strongly convex. Since the sum of a convex function and a strongly convex function is strongly convex, $F_S(w)$ itself is $\lambda\alpha$ -strongly convex.

Let $w_S \in \arg \min_{w \in W} F_S(w)$ be the exact minimizer over the convex domain W . By the definition of strong convexity, for any arbitrary $w \in W$, the function satisfies:

$$F_S(w) \geq F_S(w_S) + \langle \nabla F_S(w_S), w - w_S \rangle + \frac{\lambda\alpha}{2} \|w - w_S\|^2$$

Because w_S is the minimizer over the convex set W , the first-order optimality condition dictates that moving in any feasible direction from w_S cannot decrease the objective.

Therefore, the inner product of the gradient and any feasible direction is non-negative: $\langle \nabla F_S(w_S), w - w_S \rangle \geq 0$. Substituting this gives a simpler lower bound:

$$F_S(w) \geq F_S(w_S) + \frac{\lambda\alpha}{2} \|w - w_S\|^2$$

Now, we apply this property to the approximate minimizer $\tilde{w}_S$ returned by our algorithm:

$$F_S(\tilde{w}_S) - F_S(w_S) \geq \frac{\lambda\alpha}{2} \|\tilde{w}_S - w_S\|^2$$

We are provided with a deterministic optimization guarantee that the algorithm's objective value is at most η worse than the optimal: $F_S(\tilde{w}_S) - F_S(w_S) \leq \eta$. Combining these inequalities allows us to directly bound the parameter distance:

$$\frac{\lambda\alpha}{2} \|\tilde{w}_S - w_S\|^2 \leq \eta \implies \|\tilde{w}_S - w_S\|^2 \leq \frac{2\eta}{\lambda\alpha} \implies \|\tilde{w}_S - w_S\| \leq \sqrt{\frac{2\eta}{\lambda\alpha}}$$

2. Stability of approximate RERM.

Claim: For S and S' differing in one example, $|\ell(\tilde{w}_S, z) - \ell(\tilde{w}_{S'}, z)| \leq \frac{2G^2}{\lambda\alpha n} + 2G\sqrt{\frac{2\eta}{\lambda\alpha}}$

Proof:

We want to bound the difference in loss for our approximate RERM when trained on two samples S and S' that differ by exactly one point. First, we will bound the parameter distance between the two approximate minimizers, $\tilde{w}_S$ and $\tilde{w}_{S'}$, using the triangle inequality for the parameter norm:

$$\begin{aligned}\|\tilde{w}_S - \tilde{w}_{S'}\| &= \|\tilde{w}_S - w_S + w_S - w_{S'} + w_{S'} - \tilde{w}_{S'}\| \\ \|\tilde{w}_S - \tilde{w}_{S'}\| &\leq \|\tilde{w}_S - w_S\| + \|w_S - w_{S'}\| + \|w_{S'} - \tilde{w}_{S'}\|\end{aligned}$$

We have bounds for each of these three terms:

1. From Part 1, the distance from the approximate minimizer on S to the exact minimizer on S is $\|\tilde{w}_S - w_S\| \leq \sqrt{\frac{2\eta}{\lambda\alpha}}$.
2. Similarly, on sample S' , the algorithm guarantees $\|\tilde{w}_{S'} - w_{S'}\| \leq \sqrt{\frac{2\eta}{\lambda\alpha}}$.
3. We are given the exact RERM parameter movement bound for uniform replacement stability: because S and S' differ in one example, the exact minimizers shift by at most $\|w_S - w_{S'}\| \leq \frac{2G}{\lambda an}$.

Summing these bounds gives the total parameter movement for the approximate RERM:

$$\|\tilde{w}_S - \tilde{w}_{S'}\| \leq 2\sqrt{\frac{2\eta}{\lambda\alpha}} + \frac{2G}{\lambda an}$$

The loss function $\ell(w, z)$ is assumed to be G -Lipschitz with respect to the norm $\|\cdot\|$. This Lipschitz property allows us to translate the parameter distance directly into a bound on the loss difference for any test point z :

$$|\ell(\tilde{w}_S, z) - \ell(\tilde{w}_{S'}, z)| \leq G\|\tilde{w}_S - \tilde{w}_{S'}\|$$

Substituting our established bound for the parameter distance:

$$|\ell(\tilde{w}_S, z) - \ell(\tilde{w}_{S'}, z)| \leq G\left(\frac{2G}{\lambda an} + 2\sqrt{\frac{2\eta}{\lambda\alpha}}\right) = \frac{2G^2}{\lambda an} + 2G\sqrt{\frac{2\eta}{\lambda\alpha}}$$

3. Learning guarantee with optimization error.

Claim: $E_S L_D(\tilde{w}_S) \leq L_D(u) + \lambda\Psi(u) + \frac{2G^2}{\lambda an} + G\sqrt{\frac{2\eta}{\lambda\alpha}}$

Proof:

To establish a risk bound for the approximate minimizer, we relate its true risk back to the true risk of the exact minimizer. For any single test point z , using the G -Lipschitz property of

the loss function ℓ , the loss of the approximate minimizer is bounded by the loss of the exact minimizer plus a penalty for the distance between them:

$$\begin{aligned}\ell(\tilde{w}_S, z) - \ell(w_S, z) &\leq |\ell(\tilde{w}_S, z) - \ell(w_S, z)| \leq G\|\tilde{w}_S - w_S\| \\ \ell(\tilde{w}_S, z) &\leq \ell(w_S, z) + G\|\tilde{w}_S - w_S\|\end{aligned}$$

The true risk $L_D(w)$ is simply the expectation of the loss over $z \sim D$. Taking the expectation of both sides over z gives:

$$L_D(\tilde{w}_S) \leq L_D(w_S) + G\|\tilde{w}_S - w_S\|$$

Next, we want to find the expected true risk by taking the expectation over the random draw of the training sample $S \sim D^n$. By linearity of expectation:

$$\mathbb{E}_S L_D(\tilde{w}_S) \leq \mathbb{E}_S L_D(w_S) + G \mathbb{E}_S \|\tilde{w}_S - w_S\|$$

From Part 1, we proved a deterministic bound that holds for every sample S : $\|\tilde{w}_S - w_S\| \leq \sqrt{\frac{2\eta}{\lambda\alpha}}$

. We can substitute this worst-case bound into the expectation:

$$\mathbb{E}_S L_D(\tilde{w}_S) \leq \mathbb{E}_S L_D(w_S) + G\sqrt{\frac{2\eta}{\lambda\alpha}}$$

Finally, we recognize that $\mathbb{E}_S L_D(w_S)$ is exactly the expected risk of the exact RERM rule.

According to the Learning from Stable RERM theorem, for any comparator $u \in W$, this exact RERM risk is bounded by:

$$\mathbb{E}_S L_D(w_S) \leq L_D(u) + \lambda\Psi(u) + \frac{2G^2}{\lambda\alpha n}$$

Substituting this RERM bound into our inequality gives the final expected true-risk bound for the approximate minimizer:

$$\mathbb{E}_S L_D(\tilde{w}_S) \leq L_D(u) + \lambda\Psi(u) + \frac{2G^2}{\lambda\alpha n} + G\sqrt{\frac{2\eta}{\lambda\alpha}}$$

4. How accurate must the optimizer be?

Claim: A sufficient condition on η is $\eta \leq \frac{GB}{\sqrt{2\alpha n^{3/2}}}$, which yields an excess-risk bound of

$$L_D(u) + \frac{(2\sqrt{2}+1)GB}{\sqrt{\alpha n}}.$$

Proof:

We are given a specific optimal scaling for the regularization parameter: $\lambda = \sqrt{\frac{2G^2}{aB^2n}} = \frac{\sqrt{2}G}{B\sqrt{an}}$.

We want to compete with all comparators u satisfying $\Psi(u) \leq B^2$.

First, let's simplify the first three terms of our bound from Part 3 under these conditions:

$$L_D(u) + \lambda\Psi(u) + \frac{2G^2}{\lambda an}$$

Since $\Psi(u) \leq B^2$, we can upper bound $\lambda\Psi(u)$ by λB^2 . Substituting our specific choice of λ :

$$\begin{aligned} \lambda B^2 + \frac{2G^2}{\lambda an} &= \left(\frac{\sqrt{2}G}{B\sqrt{an}} \right) B^2 + \frac{2G^2}{\left(\frac{\sqrt{2}G}{B\sqrt{an}} \right) an} \\ &= \frac{\sqrt{2}GB}{\sqrt{an}} + \frac{2G^2 \cdot B\sqrt{an}}{\sqrt{2}Gan} = \frac{\sqrt{2}GB}{\sqrt{an}} + \frac{\sqrt{2}GB}{\sqrt{an}} = \frac{2\sqrt{2}GB}{\sqrt{an}} \end{aligned}$$

So the base exact RERM bound is $L_D(u) + \frac{2\sqrt{2}GB}{\sqrt{an}}$.

Now, we are asked to find a sufficient condition on the optimization accuracy η such that the optimization error term is bounded by $\frac{GB}{\sqrt{an}}$:

$$G\sqrt{\frac{2\eta}{\lambda\alpha}} \leq \frac{GB}{\sqrt{an}}$$

Dividing by G and squaring both sides:

$$\frac{2\eta}{\lambda\alpha} \leq \frac{B^2}{an}$$

Multiplying by $\lambda\alpha/2$:

$$\eta \leq \frac{\lambda B^2}{2n}$$

To make this condition explicit in terms of fundamental parameters, we substitute the definition of λ back in:

$$\eta \leq \left(\frac{\sqrt{2}G}{B\sqrt{an}} \right) \frac{B^2}{2n} = \frac{\sqrt{2}GB^2}{2Bn\sqrt{an}} = \frac{GB}{\sqrt{2an}^{3/2}}$$

This is our sufficient condition for η . If the optimizer is this accurate, the penalty introduced by approximation is well-controlled.

Under this condition, the total excess-risk bound simply adds the $\frac{GB}{\sqrt{an}}$ penalty to the simplified exact RERM bound:

$$E_S L_D(\tilde{w}_S) \leq L_D(u) + \frac{2\sqrt{2}GB}{\sqrt{an}} + \frac{GB}{\sqrt{an}} = L_D(u) + \frac{(2\sqrt{2} + 1)GB}{\sqrt{an}}$$

5. Soft-margin linear classification.

Claim: The approximate-RERM bound is $L_D(u) + \frac{2RB}{\sqrt{n}} + \sqrt{RB\eta\sqrt{n}}$, and a sufficient condition on η is $\eta \leq \frac{RB}{n^{3/2}}$.

Proof:

For soft-margin SVM, the loss function is the hinge loss: $\ell(w, (x, y)) = (1 - y\langle w, x \rangle)_+$. For feature vectors bounded by $\|x\|_2 \leq R$, the hinge loss is known to be G -Lipschitz with $G = R$ in the Euclidean norm $\|\cdot\|_2$.

The regularizer is Euclidean weight decay: $\Psi(w) = \frac{1}{2}\|w\|_2^2$. The function $\frac{1}{2}\|w\|_2^2$ is exactly 1-strongly convex with respect to $\|\cdot\|_2$, so the strong convexity parameter is $\alpha = 1$.

We want to compete with a comparator u bounded by $\|u\|_2 \leq B$. This implies a bound on the regularizer: $\Psi(u) = \frac{1}{2}\|u\|_2^2 \leq \frac{1}{2}B^2$. In the context of Part 4, our new "B-squared" term is actually $\tilde{B}^2 = \frac{B^2}{2}$.

Let's specialize the optimal λ from Part 4 using these specific values ($G = R$, $\alpha = 1$, and target scale $\frac{B^2}{2}$):

$$\lambda = \sqrt{\frac{2R^2}{1 \cdot \left(\frac{B^2}{2}\right) \cdot n}} = \sqrt{\frac{4R^2}{B^2 n}} = \frac{2R}{B\sqrt{n}}$$

Now, substituting these into the three exact RERM terms from Part 3:

$$\lambda \Psi(u) + \frac{2G^2}{\lambda an} \leq \left(\frac{2R}{B\sqrt{n}}\right) \left(\frac{B^2}{2}\right) + \frac{2R^2}{\left(\frac{2R}{B\sqrt{n}}\right) \cdot 1 \cdot n} = \frac{RB}{\sqrt{n}} + \frac{2R^2 B \sqrt{n}}{2Rn} = \frac{RB}{\sqrt{n}} + \frac{RB}{\sqrt{n}} = \frac{2RB}{\sqrt{n}}$$

Next, let's specialize the optimization-error term $G\sqrt{\frac{2\eta}{\lambda\alpha}}$:

$$G\sqrt{\frac{2\eta}{\lambda\alpha}} = R\sqrt{\frac{2\eta}{\left(\frac{2R}{B\sqrt{n}}\right) \cdot 1}} = R\sqrt{\frac{2\eta B\sqrt{n}}{2R}} = R\sqrt{\frac{\eta B\sqrt{n}}{R}} = \sqrt{RB\eta\sqrt{n}}$$

Combining these gives the specialized approximate-RERM bound in terms of R, B, n , and η :

$$E_S L_D(\tilde{w}_S) \leq L_D(u) + \frac{2RB}{\sqrt{n}} + \sqrt{RB\eta\sqrt{n}}$$

Finally, we find the sufficient condition on η to ensure this optimization-error term is no larger than $\frac{RB}{\sqrt{n}}$. Setting up the inequality:

$$\sqrt{RB\eta\sqrt{n}} \leq \frac{RB}{\sqrt{n}}$$

Squaring both sides:

$$RB\eta\sqrt{n} \leq \frac{R^2 B^2}{n}$$

Solving for η :

$$\eta \leq \frac{R^2 B^2}{RBn\sqrt{n}} = \frac{RB}{n^{3/2}}$$

Thus, $\eta \leq \frac{RB}{n^{3/2}}$ is the sufficient condition for soft-margin linear classification to maintain optimal statistical rates despite optimization errors.

Part C: A Weighted Euclidean Geometry

1. Geometry quantities.

Claim: $\|\varphi(x)\|_{Q,*}^2 \leq \sum_{j=1}^d \frac{r_j^2}{q_j}$ and $\sup_{w \in C_b} \Psi_Q(w) = \frac{1}{2} \sum_{j=1}^d q_j b_j^2$.

Proof:

Let's tackle the dual norm bound first. The dual norm of the weighted Euclidean norm $\|v\|_Q = \sqrt{v^\top Q v}$ is defined algebraically as $\|v\|_{Q,*} = \sqrt{v^\top Q^{-1} v}$.

Since our matrix Q is explicitly given as a diagonal matrix $Q = \text{diag}(q_1, \dots, q_d)$ with strictly positive diagonal entries $q_j > 0$, its inverse Q^{-1} is simply the diagonal matrix of the reciprocals: $Q^{-1} = \text{diag}(1/q_1, \dots, 1/q_d)$.

Therefore, for the feature vector $\varphi(x)$, the squared dual norm expands as a sum of scaled squared components:

$$\|\varphi(x)\|_{Q,*}^2 = \varphi(x)^\top Q^{-1} \varphi(x) = \sum_{j=1}^d \frac{\varphi_j(x)^2}{q_j}$$

We are given the component-wise feature bounds $|\varphi_j(x)| \leq r_j$ for all x and j . Squaring this inequality gives $\varphi_j(x)^2 \leq r_j^2$. Since $q_j > 0$, dividing by q_j preserves the inequality, allowing us to upper bound the sum:

$$\|\varphi(x)\|_{Q,*}^2 \leq \sum_{j=1}^d \frac{r_j^2}{q_j}$$

Now, let's find the supremum of the regularizer. The regularizer is defined as $\Psi_Q(w) = \frac{1}{2} w^\top Q w$. Since Q is diagonal, this is just a weighted sum of squares:

$$\Psi_Q(w) = \frac{1}{2} \sum_{j=1}^d q_j w_j^2$$

We want to find the maximum of this function over the constrained domain $C_b = \{w \in \mathbb{R}^d : |w_j| \leq b_j \text{ for all } j\}$. The constraints $|w_j| \leq b_j$ imply $w_j^2 \leq b_j^2$. Because all weights q_j are strictly positive, the sum is maximized when each individual term w_j^2 takes its maximum possible value, which is b_j^2 .

Substituting $w_j^2 = b_j^2$ yields the supremum:

$$\sup_{w \in C_b} \Psi_Q(w) = \frac{1}{2} \sum_{j=1}^d q_j b_j^2$$

2. Fixed Q .

$$\textbf{Claim: } EL_D(A_\lambda(S)) \leq \inf_{w \in C_b} L_D(w) + \frac{2g}{\sqrt{n}} \sqrt{\left(\sum_{j=1}^d q_j b_j^2 \right) \left(\sum_{j=1}^d \frac{r_j^2}{q_j} \right)}$$

Proof:

We are operating with a linear predictor where the scalar loss is g -Lipschitz. We first need to establish the Lipschitz constant G of the loss function $\ell(w, z)$ with respect to our chosen parameter norm $\|\cdot\|_Q$.

Using Hölder's inequality (which states $|\langle u, v \rangle| \leq \|u\|_Q \|v\|_*$), we can bound the difference in loss for two parameter vectors w and w' :

$$|\ell(w, z) - \ell(w', z)| \leq g |\langle w - w', \varphi(x) \rangle| \leq g \|w - w'\|_Q \|\varphi(x)\|_{Q,*}$$

The true Lipschitz constant G with respect to $\|\cdot\|_Q$ is therefore $G = g \sup_x \|\varphi(x)\|_{Q,*}$. Using our result from Part 1, we can upper bound G^2 :

$$G^2 \leq g^2 \sum_{j=1}^d \frac{r_j^2}{q_j}$$

The regularizer $\Psi_Q(w) = \frac{1}{2} w^\top Q w$ is perfectly tailored to be 1-strongly convex with respect to the norm $\|w\|_Q = \sqrt{w^\top Q w}$, which means our strong convexity parameter is $\alpha = 1$.

Now, we apply the Week 8 Learning from Stable RERM theorem. For any comparator $w \in C_b$, the expected risk of the algorithm $A_\lambda(S)$ is bounded by:

$$EL_D(A_\lambda(S)) \leq L_D(w) + \lambda \Psi_Q(w) + \frac{2G^2}{\lambda \alpha n}$$

To make this bound independent of a specific w , we can bound the regularizer term using the supremum found in Part 1, $\Psi_Q(w) \leq \frac{1}{2} \sum_j q_j b_j^2$. Substituting this and our bound for G^2 (with $\alpha = 1$) gives:

$$EL_D(A_\lambda(S)) \leq L_D(w) + \frac{\lambda}{2} \left(\sum_{j=1}^d q_j b_j^2 \right) + \frac{2g^2}{\lambda n} \left(\sum_{j=1}^d \frac{r_j^2}{q_j} \right)$$

Since this holds for any $w \in C_b$, we can take the infimum over $w \in C_b$.

Finally, we optimize this bound over our choice of the regularization strength $\lambda > 0$. The expression has the form $A\lambda + B/\lambda$, which is minimized when the two terms are equal ($A\lambda = B/\lambda \implies \lambda = \sqrt{B/A}$). The minimum value is $2\sqrt{AB}$. In our case, $A = \frac{1}{2} \sum_{j=1}^d q_j b_j^2$ and $B = \frac{2g^2}{n} \sum_{j=1}^d \frac{r_j^2}{q_j}$. The minimized excess term is:

$$2 \sqrt{\left(\frac{1}{2} \sum_{j=1}^d q_j b_j^2 \right) \left(\frac{2g^2}{n} \sum_{j=1}^d \frac{r_j^2}{q_j} \right)} = 2 \sqrt{\frac{g^2}{n} \left(\sum_{j=1}^d q_j b_j^2 \right) \left(\sum_{j=1}^d \frac{r_j^2}{q_j} \right)} = \frac{2g}{\sqrt{n}} \sqrt{\left(\sum_{j=1}^d q_j b_j^2 \right) \left(\sum_{j=1}^d \frac{r_j^2}{q_j} \right)}$$

Substituting this optimized excess term back into the inequality gives the final result:

$$EL_D(A_\lambda(S)) \leq \inf_{w \in C_b} L_D(w) + \frac{2g}{\sqrt{n}} \sqrt{\left(\sum_{j=1}^d q_j b_j^2 \right) \left(\sum_{j=1}^d \frac{r_j^2}{q_j} \right)}$$

3. Best diagonal geometry.

Claim: The best excess term is $\frac{2g}{\sqrt{n}} \sum_{j=1}^d b_j r_j$.

Proof:

From Part 2, the excess risk term depends on the specific diagonal geometry Q through the product inside the square root:

$$\text{Excess Risk} \propto \sqrt{\left(\sum_{j=1}^d q_j b_j^2 \right) \left(\sum_{j=1}^d \frac{r_j^2}{q_j} \right)}$$

Our goal is to choose the diagonal elements $q_j > 0$ to minimize this product, thereby obtaining the tightest possible bound across all diagonal geometries.

We can apply the Cauchy-Schwarz inequality, which states that for any real sequences u_j and v_j , $(\sum u_j^2)(\sum v_j^2) \geq (\sum u_j v_j)^2$. Let us define our sequences strategically to match our product

terms: Let $u_j = \sqrt{q_j} b_j$ and let $v_j = \frac{r_j}{\sqrt{q_j}}$. Note that $u_j^2 = q_j b_j^2$ and $v_j^2 = \frac{r_j^2}{q_j}$.

Applying Cauchy-Schwarz:

$$\begin{aligned} \left(\sum_{j=1}^d u_j^2 \right) \left(\sum_{j=1}^d v_j^2 \right) &\geq \left(\sum_{j=1}^d u_j v_j \right)^2 \\ \left(\sum_{j=1}^d q_j b_j^2 \right) \left(\sum_{j=1}^d \frac{r_j^2}{q_j} \right) &\geq \left(\sum_{j=1}^d \left(\sqrt{q_j} b_j \right) \left(\frac{r_j}{\sqrt{q_j}} \right) \right)^2 = \left(\sum_{j=1}^d b_j r_j \right)^2 \end{aligned}$$

This establishes a fundamental lower bound on the product that is entirely independent of our choice of Q .

Furthermore, Cauchy-Schwarz tells us that this lower bound is strictly achievable (equality holds) if and only if the sequences are proportional, i.e., $u_j \propto v_j$ for all j . This condition means:

$$\sqrt{q_j} b_j \propto \frac{r_j}{\sqrt{q_j}}$$

Multiplying both sides by $\sqrt{q_j}$ gives:

$$q_j b_j \propto r_j \implies q_j \propto \frac{r_j}{b_j}$$

Since we need $q_j > 0$, setting $q_j = \frac{r_j}{b_j}$ (or any positive scalar multiple) is a valid choice that perfectly achieves the lower bound.

Substituting this achievable minimum value back into the original excess risk expression:

$$\text{Best Excess Risk} = \frac{2g}{\sqrt{n}} \sqrt{\left(\sum_{j=1}^d b_j r_j \right)^2} = \frac{2g}{\sqrt{n}} \sum_{j=1}^d b_j r_j$$

This represents the absolute best-case bound achievable by tuning the diagonal geometry Q to match the scales of the parameter domain b and feature space r .

Part D: AI Usage Report

1. This problem set was completed using a standard agentic orchestration workflow with Gemini 3.1 Pro. It generated and analyzed new proofs. SKILL.md did an excellent job of standardizing those proofs and making them easier to read; without it, the Gemini made some assumptions that were hard to follow. In this week's changelog:
 - Added updates to SKILL.md to guide Gemini to be less hand-wavy when listing non-obvious algebraic steps.
 - Integrated LLM-as-a-judge checks using a smaller Gemini model in a separate session to surface non-obvious assumptions and clarify them.
 - Updated PLAN.md with new lecture proofs and clarified ambiguous assumptions.
 - When a QED relied on unstated assumptions, I added explicit notation clarifications to PLAN.md.
 - Limited agent orchestration builds to three rounds to save on tokens.
2. I accepted Gemini's suggestions for all ERM proofs. I believe the model has received sufficient training in optimization to avoid memorizing the training data and instead generalize to the unseen examples in this problem set. In line with ERM theory, since the training loss cannot go below zero, adding more parameters (these questions) creates additional ways to fit finite data in the modern interpolating regime; assuming Gemini models behave consistently with double-descent risk theory.

3. The first set of optimization and ERM proofs contained significant jumps in logic and lacked detailed algebraic manipulation, so it was rejected. Although the QED was sensible, I wanted to refine the proofs further, so we applied the same skill used on the previous problem set. That was still too hand-wavy, so I revised the proof-writing approach again to emphasize explicitly listing non-obvious algebraic steps. On the third round of agent orchestration, the proofs presented clear, easy-to-follow algebraic steps and included the correct ERM theory in `PLAN.md`. I also ensured the agent cited the proper definitions of commonly used probability lemmas to verify the meaning of certain terms in optimization theory.
4. I reviewed the proofs myself using techniques from DSC 211, DSC 240, and DSC 241. I also populated `PLAN.md` with all relevant lecture proofs and used LLM-as-a-judge workflows to verify each derivation against both explicit and implicit assumptions. Where assumptions were too vague, I noted them in `PLAN.md` for the agent to refine the proofs further using `SKILL.md`.